Saturday 14 November | 09:30–10:00

On Graphical Freeness for Graded Linear Categories

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In this talk, I sketch a quotient-free graphical calculus for a free graded linear category over an arbitrary category, together with its application to a coherence problem (if time permits).

Linear categories [1] elegantly axiomatise a canonical interplay between linear and nonlinear maps recurring across logic and mathematics. Despite the intricate interaction of the comonad with the symmetric monoidal and the cartesian structures of a linear category (e.g. its standard formal theory comes with seventy knotty formal equations), I have established a *graphical calculus* for a free linear category over an arbitrary category [2]. To my knowledge, it is the first *quotient-free* presentation of the free linear category that *reduces algebra to geometry*: two morphisms are equal if and only if their graphical representatives are the same graph.

On the one hand, the graphical calculus induces quantitative information through an interpretation of paths as resource-sensitive computations. On the other hand, linear categories *graded* by semiring elements [3] refine linear categories, making the quantitative intuition behind them explicit. I then conjectured that the two quantitative structures coincide: the graphical calculus forms a free graded linear category over the base category. A positive answer to this question justifies graded linear categories as a *canonical* quantitative refinement of linear categories. I have obtained a proof sketch that solves the conjecture in the positive. In this talk, I introduce the graphical calculus, outline the proof, and, should time allow, state a necessary, sufficient graphical condition for *coherence* on graded linear categories.

Keywords: graded linear categories; graphical calculi; coherence
Subject area: graphical calculi
MSC2020: 18A50

References

[1] Maietti, M.E., Maneggia, P., De Paiva, V. and Ritter, E., 2005. *Relating Categorical Semantics for Intuitionistic Linear Logic*. Applied Categorical Structures, 13(1), pp.1-36.

[2] Yamada, Norihiro. *Graphical Freeness and Coherence for Linear Categories*. Available at https://sites.google.com/view/norihiro-yamada/publications.

[3] Gaboardi, M., Katsumata, S.Y., Orchard, D., Breuvart, F. and Uustalu, T., 2016. *Combining Effects and Coeffects via Grading*. ACM SIGPLAN Notices, 51(9), pp.476-489.